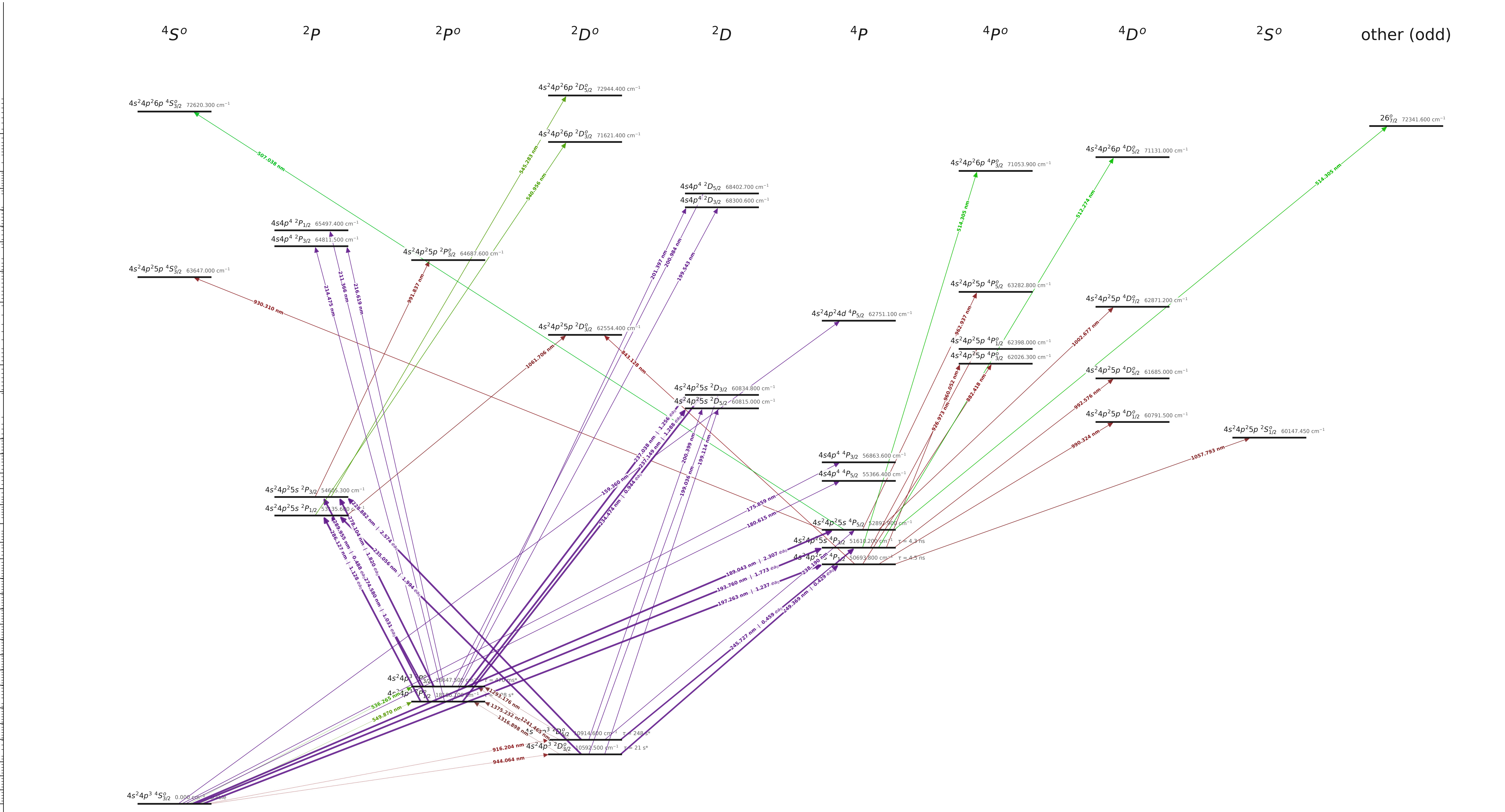


As (As I). Strongest classified transitions below 2 μm (NIST): 35 levels, 53 lines.

As⁺ ionisation limit 78950.00 cm⁻¹ = 9.78855 eV (126.662 nm)

Arrow label: vacuum wavelength | reduced dipole matrix element $\langle f || er || i \rangle$ in ea_0
 Matrix elements are measured or NIST values; * = high-accuracy theory, $\approx$ = model calculation.
 Arrow width grows with the Einstein coefficient A. Dotted: forbidden (clock, M1, E2). Dashed: Rydberg excitation, $n = 70$
 Level caption: NIST energy and measured lifetime (where one exists).



Transition	λ vac (nm)	λ air (nm)	ν (THz)	$ \langle J \mathbf{er} J'\rangle $ (ea0)	A (s^{-1})	$\Gamma/2\pi$ up. (MHz)	branch (%)	Use
$4s^2 4p^3 \ ^4S_{3/2}^o - 4s^2 4p^2 5s \ ^4P_{3/2}$	193.7601	-	1547.2349	1.773	2.190e+08	37.01	94.2	resonance line
$4s^2 4p^3 \ ^4S_{3/2}^o - 4s^2 4p^2 5s \ ^4P_{1/2}$	197.2628	-	1519.7619	1.237	2.020e+08	35.37	90.9	resonance line
$4s^2 4p^3 \ ^4S_{3/2}^o - 4s^2 4p^3 \ ^2P_{3/2}^o$	536.2649	536.1158	559.0380	-	1.600e+00	3.386e-07	75.2	forbidden line within the ground configuration
$4s^2 4p^3 \ ^4S_{3/2}^o - 4s^2 4p^3 \ ^2P_{1/2}^o$	549.8705	549.7178	545.2056	-	6.910e-01	1.474e-07	74.6	forbidden line within the ground configuration
$4s^2 4p^3 \ ^4S_{3/2}^o - 4s^2 4p^3 \ ^2D_{5/2}^o$	916.2040	915.9526	327.2115	-	5.300e-03	6.418e-10	100	forbidden line within the ground configuration
$4s^2 4p^3 \ ^4S_{3/2}^o - 4s^2 4p^3 \ ^2D_{3/2}^o$	944.0642	943.8053	317.5552	-	7.490e-02	7.579e-09	100	forbidden line within the ground configuration
$4s^2 4p^3 \ ^2D_{5/2}^o - 4s^2 4p^3 \ ^2P_{3/2}^o$	1241.4649	1241.1253	241.4828	-	6.720e-01	3.386e-07	31.6	forbidden line within the ground configuration
$4s^2 4p^3 \ ^2D_{3/2}^o - 4s^2 4p^3 \ ^2P_{3/2}^o$	1293.1759	1292.8223	231.8265	-	4.400e-01	3.386e-07	20.7	forbidden line within the ground configuration
$4s^2 4p^3 \ ^2D_{5/2}^o - 4s^2 4p^3 \ ^2P_{1/2}^o$	1316.8984	1316.5384	227.6504	-	4.200e-01	1.474e-07	45.4	forbidden line within the ground configuration
$4s^2 4p^3 \ ^2D_{3/2}^o - 4s^2 4p^3 \ ^2P_{1/2}^o$	1375.2321	1374.8561	217.9941	-	5.900e-02	1.474e-07	6.37	forbidden line within the ground configuration

Level	A (MHz)	B (MHz)	Shift of each F level from the centre of gravity (MHz)			
$4s^2 4p^3 \ ^4S_{3/2}$	-66.2040(10)	-0.535(3)	F=0: +247.60	F=1: +181.93	F=2: +50.05	F=3: -149.09
$4s^2 4p^3 \ ^2D_{3/2}$	480(9)	-	F=0: -1800.00	F=1: -1320.00	F=2: -360.00	F=3: +1080.00
$4s^2 4p^3 \ ^2D_{5/2}$	725(26)	-	F=1: -3806.25	F=2: -2356.25	F=3: -181.25	F=4: +2718.75
$4s^2 4p^3 \ ^2P_{1/2}$	1909(14)	-	F=1: -2386.25	F=2: +1431.75		
$4s^2 4p^3 \ ^2P_{3/2}$	359(59)	-	F=0: -1346.25	F=1: -987.25	F=2: -269.25	F=3: +807.75

Measured A, B constants; sources in the data file.